MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

02

10 answers · Rationale-rich · Teach from doc

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Q1

A

A. $T = \frac{mv^2}{L}$

Choice A is correct. To write the formula as T in terms of m, v, and L means to isolate T on one side of the equation. First, multiply both sides of the equation by m, which gives $mv^2 = \frac{mLT}{m}$, which simplifies to $mv^2 = LT$. Next, divide both sides of the equation by L, which gives $\frac{mv^2}{L} = \frac{LT}{L}$, which simplifies to $T = \frac{mv^2}{L}$.

Choices B, C, and D are incorrect and may be the result of incorrectly applying operations to each side of the equation.

Q2

6

The correct answer is 6. It's given that $y = 4x$ and $y = x^2 - 12$. Since $y = 4x$, substituting $4x$ for y in the second equation of the given system yields $4x = x^2 - 12$. Subtracting $4x$ from both sides of this equation yields $0 = x^2 - 4x - 12$. This equation can be rewritten as $0 = x - 6x + 2$. By the zero product property, $x - 6 = 0$ or $x + 2 = 0$. Adding 6 to both sides of the equation $x - 6 = 0$ yields $x = 6$. Subtracting 2 from both sides of the equation $x + 2 = 0$ yields $x = -2$. Therefore, solutions to the given system of equations occur when $x = 6$ and when $x = -2$. It's given that a solution to the given system of equations is x, y, where $x > 0$. Since 6 is greater than 0, it follows that the value of x is 6.

Q3

B

B. -4

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $x + 4^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x =$ -4. Therefore, the value of x is -4.

Choice A is incorrect. This is the value of y, not x.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

either 8 or 9

The correct answer is either 8 or 9. The first equation can be rewritten as $y = 17 - x$. Substituting $17 - x$ for y in the second equation gives $x(17 - x) = 72$. By applying the distributive property, this can be rewritten as $17x - x^2 = 72$. Subtracting 72 from both sides of the equation yields $x^2 - 17x + 72 = 0$. Factoring the left-hand side of this equation yields $(x - 8)(x - 9) = 0$. Applying the Zero Product Property, it follows that $x - 8 = 0$ and $x - 9 = 0$. Solving each equation for x yields $x = 8$ and $x = 9$ respectively. Note that 8 and 9 are examples of ways to enter a correct answer.

Q5

A

A. $x + 15^2 = 0$

Choice A is correct. Taking the square root of both sides of $(x + 15)^2 = 0$ yields $x + 15 = 0$. Subtracting 15 from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Q6

-3

The correct answer is -3. Dividing both sides of the given equation by 38 yields $x^2 = 9$. Taking the square root of both sides of this equation yields the solutions $x = 3$ and $x = -3$. Therefore, the negative solution to the given equation is -3.

Q7

30

Accept also: `-30`

The correct answer is either -30 or 30. Adding 7 to each side of the given equation yields $d - 30d + 30 = 0$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $d - 30 = 0$ or $d + 30 = 0$. Adding 30 to each side of the equation $d - 30 = 0$ yields $d = 30$. Subtracting 30 from each side of the equation $d + 30 = 0$ yields $d = -30$. Therefore, the solutions to the given equation are -30 and 30. Note that -30 and 30 are examples of ways to enter a correct answer.

Q8 D

D. $C = 19 - \frac{P}{N}$

Choice D is correct. It's given that the values of P, N, and C are positive. Therefore, dividing each side of the given equation by N yields $\frac{P}{N} = 19 - C$. Subtracting 19 from each side of this equation yields $\frac{P}{N} - 19 = -C$. Dividing each side of this equation by -1 yields $19 - \frac{P}{N} = C$, or $C = 19 - \frac{P}{N}$.

Choice A is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

Choice B is incorrect. This equation is equivalent to $P = 19 - NC$, not $P = N19 - C$.

Choice C is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

Q9 5

The correct answer is 5. The given quadratic equation can be rewritten in factored form as $(x - 8)(x + 3) = 0$. Based on the zero product property, it follows that $x - 8 = 0$ or $x + 3 = 0$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. Therefore, the solutions to the given equation are 8 and -3. It follows that the sum of the solutions to the given equation is $8 + (-3)$, or 5.

Q10 A

A. $w = -150vx$

Choice A is correct. It's given that x is positive. Therefore, multiplying each side of the given equation by $-150x$ yields $-150xv = w$, which is equivalent to $w = -150vx$. Thus, the equation $w = -150vx$ correctly expresses w in terms of v and x.

Choice B is incorrect. This equation is equivalent to $v = -\frac{wx}{150}$.

Choice C is incorrect. This equation is equivalent to $v = -\frac{x}{150w}$.

Choice D is incorrect. This equation is equivalent to $v = w - 150x$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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